# BorBann
# A Real Estate Information Platform
# (Project Proposal)

by

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Faculty of Engineering
Kasetsart University
Bangkok, Thailand

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**This Project Submitted in Partial Fulfillment of the Requirements for Bachelor Degree of Engineering (Software Engineering)**

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Advisor . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . Date. . . /. . . /. . .
(Assoc.Prof.Dr. Kitsana Waiyamai)

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# Chapter 1

# Introduction

## 1.1 Background

The global real estate market shows positive growth trends despite various challenges. Real estate market remains a key sector for people like homebuyers and investors, influenced by complex characteristics with heterogeneous nature and affected by numerous elements like policies and consumer trends. The market is projected to grow from $4,143.71 billion in 2024 to $4,466.58 billion in 2025, and global real estate investment volumes continue to increase in 2024[**?**].

Thailand's real estate market faces both challenges and opportunities in the current economic climate. The real estate market in Thailand, especially in Bangkok in 2024-2025, faces challenges such as high household debt, strict credit conditions, and rising costs causing market contraction, decreasing project launches by 43.72% in 2024[**?**]. However, it also presents opportunities for recovery through foreign investment driven by tourism recovery, government incentives, and technological advancements. Condominiums remain attractive for rental yields and foreign buyers, while the housing market has moderate growth supported by infrastructure development.

Crucially, these market opportunities can significantly undermined by Thailand's real estate information ecosystem, which suffers from several structural limitations, including outdated listings, lack of official transaction data, and data fragmentation[**?**]. Addressing these issues is essential for maximizing Thailand's market potential and enabling effective decision-making for both homebuyers and investors seeking to capitalize on the available opportunities.

## 1.2 Problem Statement

Even with large amounts of real estate data available through various channels including property listings, historical transaction records, and news, investors still face challenges in aggregating, contextualizing, and deriving action from these sources. Additionally, the Thai real estate market has different characteristics from other countries because of the limited availability of official property transaction records, listing duplication across platforms, and less standardized data compared to more mature markets.

Current platforms in Thailand like DDProperty, Hipflat, and Baania lack many important contextual factors such as climate risk assessments, neighborhood-specific news, and local beliefs that influence property valuation and real estate investment in the long term.

While platforms like Zillow and House Canary have advanced and comprehensive real estate analytics, predictive analytics, and user-friendly interfaces, they do not operate in Thailand. Even if these advanced platforms were to enter the Thai market, they would face challenges adapting their algorithms to the local context. Their models are calibrated to markets with standardized property classifications, valuations that vary by developers, and consistent transaction data—elements that are limited in Thailand's real estate ecosystem.

The BorBann platform addresses these challenges and aims to help users by creating a real estate data platform integrated with artificial intelligence, geospatial analytics, and data aggregation by focusing on analytics rather than transaction facilitation.

## 1.3 Solution Overview

BorBann will function as real estate data platform to users, integrating multiple data sources with advanced analytics. Below are the features and their details.

### Borbann AI Agent (Agentic Workflow)

- **Natural Language Interface:** Chat with the system to query locations ("How is the flooding here?")
- **Tool Orchestration:** Automatically selects and executes the right tools (e.g., transit scoring, risk checking) based on user intent
- **Spatial Context Awareness:** Understands user interactions like dropping pins or drawing bounding boxes

### Site Viability & Catchment Analysis

- **Commercial Site Scoring:** Evaluate business potential (e.g., for a Cafe) based on traffic magnets vs. competitors
- **Isochrone Visualization:** Visualizing 15-minute walking/driving reach from any point
- **Livability Scores:** Composite scoring for Transit, Walkability, and Schools

## Local Contextual Analytics

- **Environmental risk assessment:** Evaluate flood risk, natural disaster vulnerability, and air quality
- **Facility proximity analysis:** Calculate accessibility to schools, hospitals, transit, and commercial centers
- **Neighborhood quality scoring:** Generate composite metrics for area evaluation

## Explainable Price Prediction Model

- **Feature importance analysis:** Quantify and rank factors influencing property prices
- **Adjustable characteristics modeling:** Adjust property characteristics to visualize price impacts
- **Confidence intervals:** Provide lower/upper price bounds for realistic expectations
- **Factor categorization:** Group influences by type (property features, location, market trends)
- **Natural language explanations:** Generate readable summaries of price determinants
- **Visual breakdowns:** Display contribution percentages and relationship graphs

## Geospatial Visualization

- **Heatmap generation:** Create density visualizations for environmental factors, pricing, and metrics
- **Geospatial analytics:** Calculate analytics for custom geographic areas

## 1.4 Target User

| User Type | Description | Needs |
|---|---|---|
| **Real Estate Investors** | Individuals focused on maximizing long-term investment, including foreign investors | • Investment analysis<br>• Supporting data for decision making |
| **Homebuyers** | First-time purchasers, residents looking to relocate within Thailand, and expats seeking housing | • Property comparisons<br>• Neighborhood insights<br>• Pricing guidance |

Table 1.1: Target Users and Their Needs

## 1.5 Benefit

The BorBann platform will provide numerous benefits to the Thai real estate market. It improves market transparency by enhancing accessibility to market information. It also helps both homebuyers and investors reduce research time, achieve lower transaction risks, and discover overlooked investment opportunities. Additionally, the platform will effectively represent the unique characteristics of the Thai real estate market.

## 1.6 Terminology

| Term | Definition |
|---|---|
| **Local Analytics** | Analysis focused on extremely specific geographic areas, such as neighborhoods or even individual streets, to provide highly relevant insights. |
| **Price Prediction Model** | An algorithm or statistical model that forecasts property values based on historical data, market trends, and various property characteristics. |
| **Proximity Analysis** | The study of spatial relationships between geographic features, typically to evaluate the distance between properties and amenities or services. |
| **Geospatial Visualization** | The graphical representation of data with a geographic or spatial component, often through maps and interactive displays. |

Table 1.2: Terminology

# Chapter 2

# Literature Review and Related Work

This chapter presents a review of existing platforms in the domain of real estate analytics and information systems. This review includes both international and Thai platforms, their features, strengths, and limitations. The analysis establishes the current state of real estate information platforms and identifies opportunities for the BorBann platform to address unmet needs in the Thai market.

## 2.1 Competitor Analysis

Many real estate platforms primarily function as property listing aggregators rather than comprehensive information systems. Their features support transaction facilitation through showcasing available properties, while offering limited analytical tools for market understanding. However, platforms like House Canary represent exceptions, operating specifically as information systems that provide investors with data analytics and market insights to support evidence-based decision-making in real estate investments.

| Feature | BorBann (Proposed) | DDProperty | Hipflat | House Canary | Zillow |
|---|---|---|---|---|---|
| **Borbann AI Agent (ReAct)** | **Yes** | No | No | No | No |
| **Site Viability & Catchment Analysis** | **Yes** | No | No | No | No |
| **Local Contextual Analytics** | **Yes** | No | No | Yes (Not optimized for Thailand) | Yes (Not optimized for Thailand) |
| **Explainable Price Prediction Model** | **Yes** | No | No | No | No |
| **Geospatial Visualization** | **Yes** | Yes | Yes | Yes | Yes |

Table 2.1: Feature Comparison: BorBann vs Other Platforms

Table **??** demonstrates BorBann's technical advantages in the real estate analytics market. While all platforms offer geospatial visualization, BorBann's implementation includes advanced analytics like climate assessment, matching international platforms but surpassing local Thai competitors. BorBann's AI Agent allows natural language interaction for complex queries, unlike traditional filter-based search. For local contextual analytics, BorBann provides Thailand-optimized insights including weather patterns and population density, whereas DDProperty/Hipflat only show basic nearby facilities, and international platforms lack Thailand-specific optimization. Most distinctively, BorBann's price prediction model prioritizes explainability and interpretability, revealing the reasoning behind valuations rather than presenting opaque predictions like competing platforms.

| **Technical Aspect** | **DDProperty** | **Hipflat** | **House Canary** | **Zillow** | **BorBann (Proposed)** |
|---|---|---|---|---|---|
| **Data Sources** | User-submitted, Proprietary | User-submitted, Proprietary | Proprietary | Multiple Sources, Proprietary | Open Data, User-submitted |
| **ML Implementation** | Basic Prediction | None | Black-box Models | Black-box Models | Explainable Models |

Table 2.2: Comprehensive Technical Comparison

Table **??** highlights key technical differences between BorBann and competing platforms. For data sources, while competitors rely heavily on government or user-submitted data, BorBann uniquely leverages open data combined with APIs to build a more comprehensive dataset. Regarding machine learning, BorBann distinguishes itself by implementing explainable models, providing transparency in its predictions. This contrasts with DDProperty's basic prediction capabilities, Hipflat's complete lack of ML features, and the black-box approaches of House Canary and Zillow where prediction logic remains hidden from users.

# Chapter 3

# Requirement Analysis

## 3.1 Stakeholder Analysis

The stakeholder landscape for BorBann includes primary stakeholders who directly interact with the platform and secondary stakeholders who indirectly influence its adoption and effectiveness. Recognizing these groups' specific needs helps develop a platform that delivers value across the real estate ecosystem.

| Stakeholder Category | Stakeholder Group | Key Interests | Primary Requirements |
|---|---|---|---|
| **Primary** | Real Estate Investors | Environment risk assessment | Advanced analytics, predictive modeling, risk scoring |
| **Primary** | Homebuyers | Affordability, area quality, lifestyle alignment, environmental factors | User-friendly interface, neighborhood insights, price comparisons, risk assessments |
| **Secondary** | Real Estate Agencies | Market positioning, client advisement, property valuation | Property data, reliable market insights |

Table 3.1: BorBann Platform Stakeholder Summary

The primary stakeholders are direct users who rely on BorBann's capabilities for informed real estate decisions. Their requirements range from investment analytics to intuitive neighborhood quality assessments. Secondary stakeholders like real estate agencies can influence platform adoption and serve as valuable data partners.

### Primary Stakeholders

Primary stakeholders are those who directly use or are significantly affected by the BorBann platform. They rely on its capabilities to make informed decisions about real estate investments, purchases, and market trends.

### Real Estate Investors

Real estate investors use BorBann to identify profitable investment, assess risks, and make data-driven decisions.

**Interests and Concerns:**

- Maximizing return on investment
- Identifying growth areas
- Risk assessment
- Diversification across neighborhoods and property classes

**Requirements:**

- Advanced analytics tools that is customizable
- Predictive price modeling with easy to understand explaination
- Risk scores in each area based on multiple factors

### Homebuyers

Homebuyers rely on BorBann to find properties that align with their budget, lifestyle, and long-term needs. The platform helps them assess affordability, neighborhood quality, and potential risks.

**Interests and Concerns:**

- Property affordability and financing options
- Area quality
- Location amenities and lifestyle alignment
- Environmental factors including flood and pollution risks
- Commute time to work, schools, and essential services
- School districts and educational quality

**Requirements:**

- Intuitive, user-friendly interface with minimal learning curve
- Comprehensive neighborhood insights including safety metrics
- Price comparison tools with historical context

- Property quality and developer reputation metrics
- Detailed flood risk and environmental quality assessment
- Transit accessibility maps with time-based visualizations
- Education quality indicators and school zone mapping

### Secondary Stakeholders

Secondary stakeholders are indirectly impacted by the BorBann platform, influencing its adoption, data availability, and overall market reach.

#### Real Estate Agencies

Real estate agencies act as intermediaries between property buyers and sellers. They use BorBann to improve their advisory services, gain a competitive edge, and provide market insights to clients.

**Interests and Concerns:**

- Market positioning relative to competitors
- Client advisement based on reliable data
- Access to comprehensive property information
- Accurate property valuation to support transactions

**Influence:**

- Potential data partners and platform promoters
- Can significantly influence adoption rates among clients
- May provide valuable transaction data not available elsewhere

## 3.2 User Stories

User stories capture the essential needs and goals of the BorBann platform's target users from their perspective. These stories follow the standard format: "As a [user type], I want to [action/feature] so that [benefit/value]."

Table 3.2: User Stories with Acceptance Criteria

| User Story | Acceptance Criteria |
|---|---|
| **Local Contextual Analytics** | |
| As a user, I can see all contextual data for specific areas to make informed decisions | • System displays at least 5 contextual data points for any selected area<br>• Historical trends for environmental factors available for at least 12 months |
| As a user, I want to see flood risk assessments for properties with historical flooding data | • Flood risk presented on a 5-point scale with historical context<br>• System shows flood history for the past 10 years when available |
| As a user, I want to see daily air quality metrics around a property with historical trends | • Air quality index displayed with daily updates<br>• Historical air quality data presented in trend graphs |
| As a user, I want to see all schools within a custom radius, including distance, ratings, and types | • System displays all schools within user-defined radius<br>• Each school listing includes distance, type, and quality rating |
| As a user, I want to view healthcare facilities near a property with distance and service information | • Healthcare facilities categorized by type (hospital, clinic, etc.)<br>• Distance and basic service information provided for each facility |
| **Explainable Price Prediction Model** | |
| As a user, I want to see how specific contextual factors influence the property's predicted price | • System displays top 5 factors influencing price with percentage contribution<br>• Visual indicators show positive/negative impact of each factor |

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Table 3.2 – Continued from previous page

| User Story | Acceptance Criteria |
|---|---|
| As a user, I want the model to be interpretable so I can understand factors affecting the price | • Plain language explanations accompany each prediction<br>• Interactive elements allow exploration of factor relationships |
| As a user, I want a statement or reason to back the prediction so I can trust the system's valuation | • Each prediction includes at least 3 specific supporting statements<br>• System indicates confidence level for each prediction |
| As a user, I want to see a predicted price range for any property I select | • System shows lower and upper bounds for predicted price |
| As a user, I want to understand how the model derives the result so I can explain the valuation to others | • System provides visual breakdown of prediction process |
| **Geospatial Visualization** | |
| As a user, I can see property listings on the map and click them to view detailed information | • Property details popup appears within 1 second of clicking a marker<br>• Popup contains at least 5 key property attributes |
| As a user, I can see sections of the same property group to identify properties from the same development | • Properties from same development visually grouped with distinct boundaries<br>• Group name appears when hovering over grouped properties |
| As a user, I can see multiple map visualization types to analyze different environmental factors | • System offers at least 5 different visualization overlays<br>• Users can toggle between visualizations without page reload |

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Table 3.2 – Continued from previous page

| User Story | Acceptance Criteria |
|---|---|
| As a user, I want to pan and zoom on an interactive property map to explore different areas efficiently | • Map responds to standard pan/zoom gestures within 100ms<br>• Property markers adjust density based on zoom level |
| As a user, I want to set a custom radius around a point on the map to analyze the surrounding area | • User can place and adjust analysis radius on any map location |
| **Site Analysis & Catchment** | |
| As a business owner, I want to see a site score for a specific category (e.g., Cafe) to evaluate commercial potential | • System provides a 0-100 score based on magnets and competition<br>• Analysis includes count of traffic drivers within 1km |
| As a user, I want to visualize the 15-minute reachable area from a location to understand accessibility | • Map displays an isochrone polygon for walking or driving<br>• Polygon overlay includes estimated population reach |
| **Borbann AI Agent** | |
| As a user, I want to ask natural language questions about properties and have the system execute the right tools to answer | • Agent breaks down complex queries into steps (e.g., search, then calculate score)<br>• UI shows the ”thought process” and tools being used |
| As a user, I want to drag and drop a pin or draw a box on the map to define the search area for the agent | • Agent recognizes the spatial context (coordinates or bounding box) from the UI action<br>• Search results are constrained to the defined area |

## 3.3 Use Case Diagram

The Use Case Diagram for the BorBann platform, shown in Figure ??, illustrates the primary interactions between the system and its three main user types: Real Estate Investors, Homebuyers, and Property Developers. The diagram captures the core functionality of the platform and how different users interact with its features.

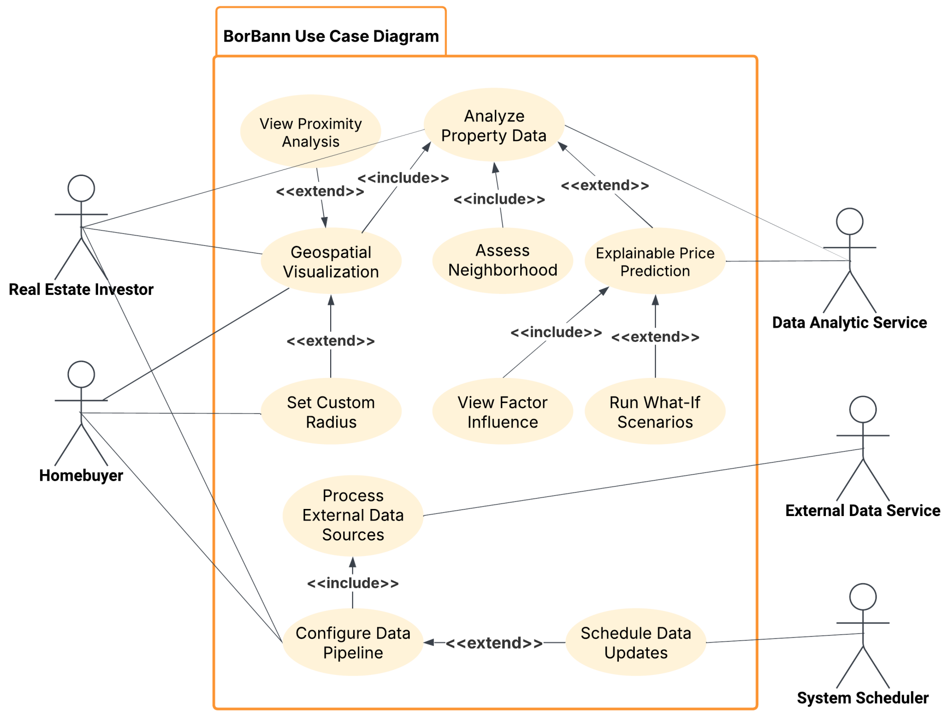


Figure 3.1: Use Case Diagram

### Primary Actors

The use case diagram identifies five key actors interacting with the BorBann system:

- **Real Estate Investor** Seeks in-depth property analytics, market trends, and investment decision support.
- **Homebuyer** Interested in residential properties, lifestyle factors, and long-term value.
- **Data Analytic Service** Provides analytic capabilities to support prediction and data analysis features.
- **External Data Service** Supplies third-party data inputs via APIs and other integrations.

- **System Scheduler** Automates routine system tasks such as scheduled data refresh operations.

## Core Use Cases

The BorBann platform provides the following primary functionalities:

- **Analyze Property Data** Allows users to examine detailed property information, metrics, and comparisons.
  - *Includes:* Assess Neighborhood
  - *Extended by:* Explainable Price Prediction
- **Geospatial Visualization** Interactive map-based visualization of property and neighborhood data.
  - *Extended by:* Set Custom Radius, View Proximity Analysis
- **Assess Neighborhood** Evaluates contextual factors around properties (e.g., schools, amenities).
  - *Includes:* View Factor Influence
  - *Extended by:* Check Environmental Risks
- **Explainable Price Prediction** AI-generated pricing insights with interpretable influencing factors.
  - *Includes:* View Factor Influence
- **Interact with AI Agent** Enables natural language queries and spatial context injection (pins/bbox) to orchestrate tools.
  - *Includes:* Site Analysis, Property Search

## Extended Functionality

The following optional use cases extend the platform's functionality:

- **View Proximity Analysis** Triggered when users want detailed proximity-based data (e.g., nearby schools, transport).
- **Set Custom Radius** Adds user-defined range settings for map-based filtering and analysis.
- **Check Environmental Risks** Enables users to access data on environmental threats (e.g., flood zones, pollution).

### Actor-System Interactions

Each actor interacts with the system in distinct ways, as shown in the diagram:

- **Real Estate Investor**
  - Initiates *Analyze Property Data*, *Geospatial Visualization*, and *Configure Data Pipeline*
  - Benefits from extended insights like *Set Custom Radius*, *View Proximity Analysis*, and *Explainable Price Prediction*
- **Homebuyer**
  - Accesses *Analyze Property Data*, *Geospatial Visualization*, *Assess Neighborhood*, and *Explainable Price Prediction*
  - Makes use of neighborhood-specific features like *View Factor Influence* and *Check Environmental Risks*
- **Data Analytic Service**
  - Collaborates with the system to enable *Explainable Price Prediction*
- **External Data Service**
  - Supports *Process External Data Sources* as part of the data ingestion workflow
- **System Scheduler**
  - Triggers *Schedule Data Updates* as an automated background process

### Use Case Relationships

- **<<include>>** Represents mandatory sub-functions:
  - *View Factor Influence* is included in both *Assess Neighborhood* and *Explainable Price Prediction*
  - *Process External Data Sources* is included in *Configure Data Pipeline*
- **<<extend>>** Represents optional or conditional behavior:
  - *Set Custom Radius* and *View Proximity Analysis* extend *Geospatial Visualization*
  - *Check Environmental Risks* extends *Assess Neighborhood*
  - *Schedule Data Updates* extends *Configure Data Pipeline*

## 3.4 Use Case Model

The Use Case Model details the interactions depicted in the Use Case Diagram, describing each use case within BorBann's core features.

| **Use Case Name** | **View Property Insights** |
|---|---|
| **Actors** | Real Estate Investor, Homebuyer, Property Developer |
| **Description** | Provides users with comprehensive analytics and contextual information about specific properties |
| **Preconditions** | User has selected a property from search results |
| **Basic Flow** | 1. User selects a property for detailed viewing<br>2. System retrieves comprehensive property data<br>3. System presents property details with analytics<br>4. System displays neighborhood characteristics<br>5. System shows historical performance metrics<br>6. User reviews information |
| **Alternative Flows** | • User can extend to save the property as a favorite<br>• User can extend to view explainable price predictions<br>• User can request additional specific analytics |
| **Postconditions** | User gains comprehensive insights about the property |
| **Associated Feature** | Local Contextual Analytics - providing trend analysis and neighborhood-specific insights |

Table 3.3: View Property Insights Use Case

| Use Case Name | Explainable Price Prediction |
|---|---|
| **Actors** | Real Estate Investor, Homebuyer |
| **Description** | Provides transparent, interpretable price predictions with detailed explanations of contributing factors |
| **Preconditions** | User is viewing property insights |
| **Basic Flow** | 1. User requests price prediction for a property<br>2. System calculates predicted price using its models<br>3. System generates explanation of factors influencing the prediction<br>4. System presents prediction with confidence interval<br>5. System displays factor weights and their impact<br>6. User reviews prediction and explanation |
| **Alternative Flows** | • User can adjust factors to see impact on prediction<br>• User can compare prediction with market averages<br>• User can save prediction for future reference |
| **Postconditions** | User understands the predicted price and the reasoning behind it |
| **Associated Feature** | Explainable Price Prediction Model - providing transparent explanations for price predictions |

Table 3.4: Explainable Price Prediction Use Case

## 3.5 User Interface Design

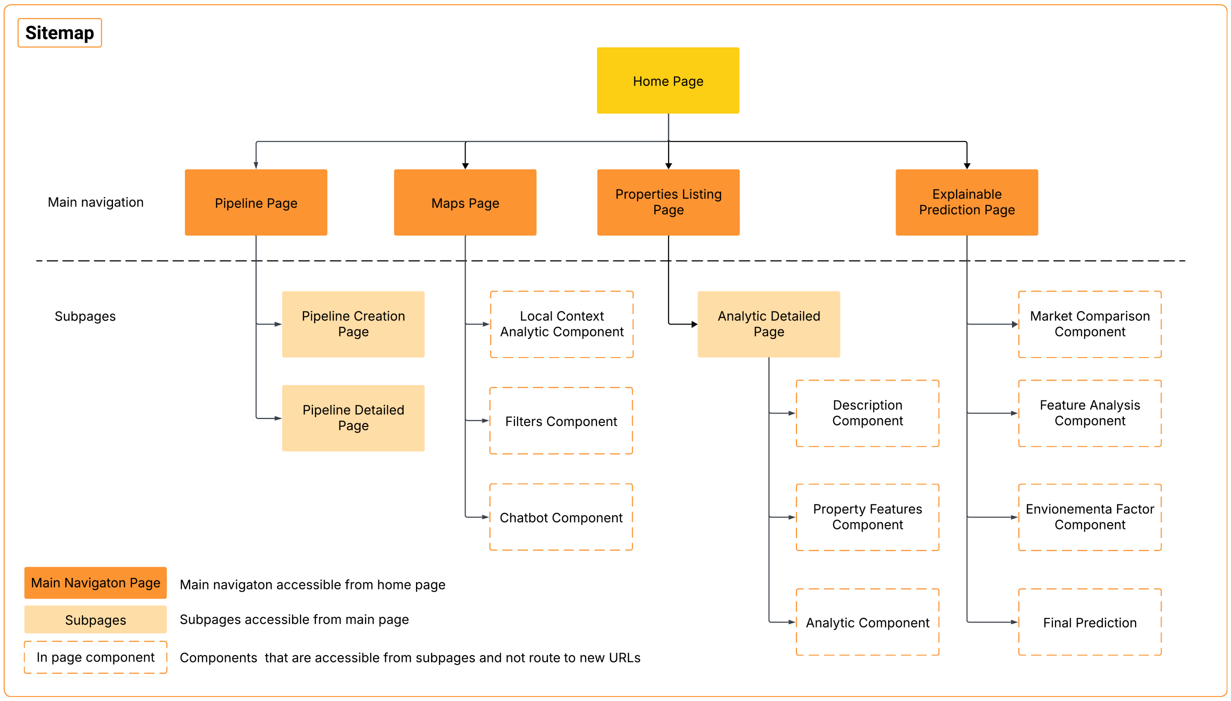


Figure 3.2: Sitemap - Platform Structure Overview

Figure ?? shows the structure of the BorBann platform. The Home Page acts as the central access point leading to four main sections: Pipeline, Maps, Properties Listing, and Explainable Prediction pages.

Each main section connects to specific subpages. The Pipeline section includes Creation and Detailed pages. The Maps section features Local Context Analytics, Filters, and Chatbot components. The Properties section provides Analytic Detailed pages with Description, Property Features, and Analytics components. The Explainable Prediction section offers Market Comparison, Feature Analysis, Environmental Factor analysis, and Final Prediction components.

This structure organizes the platform's key functions in a logical flow, making it easy for users to navigate between data management, visualization, and prediction features.

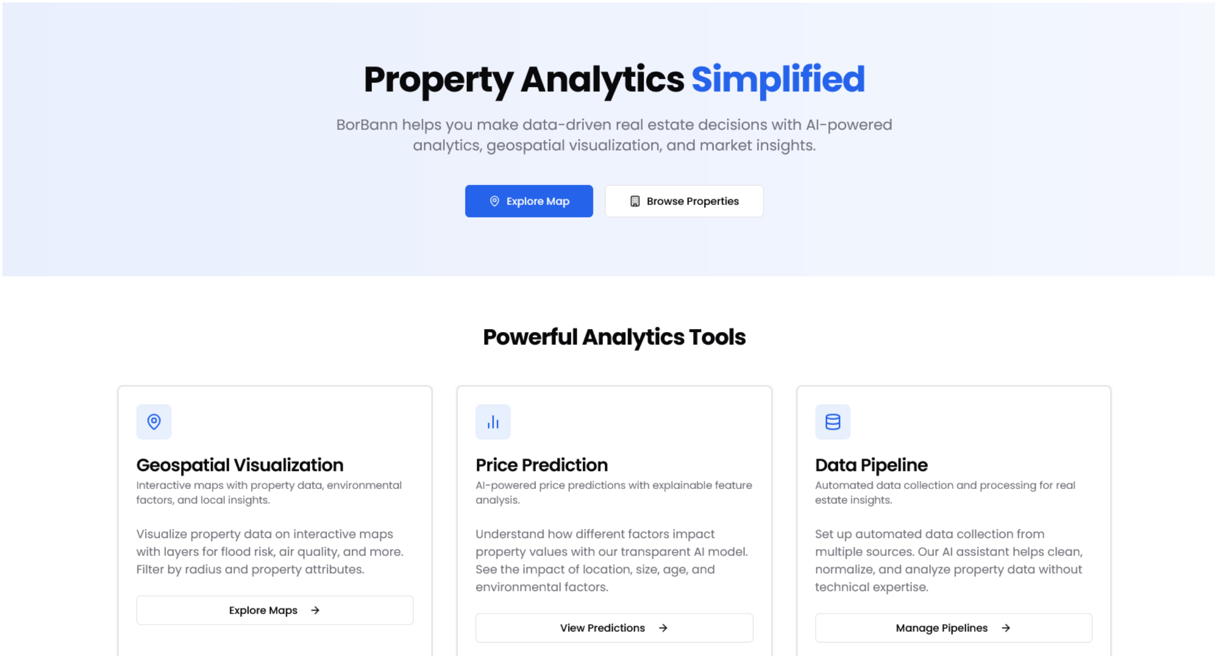


Figure 3.3: BorBann Platform Homepage

Figure ?? shows the BorBann platform homepage, which provides an intuitive entry point to the system. The page presents the core value proposition with "Property Analytics Simplified" and briefly explains how BorBann helps users make data-driven real estate decisions through analytics, geospatial visualization, and market insights. Two primary action buttons - "Explore Map" and "Browse Properties" - enable quick access to key functionality. The lower section showcases three main analytics tools: Geospatial Visualization for interactive property mapping, Price Prediction with explainable AI features, and Data Pipeline for automated data collection and processing.

Figure ?? displays the interactive property map with color-coded markers and navigation controls. The interface supports pan/zoom gestures and provides filtering options through the sidebar.

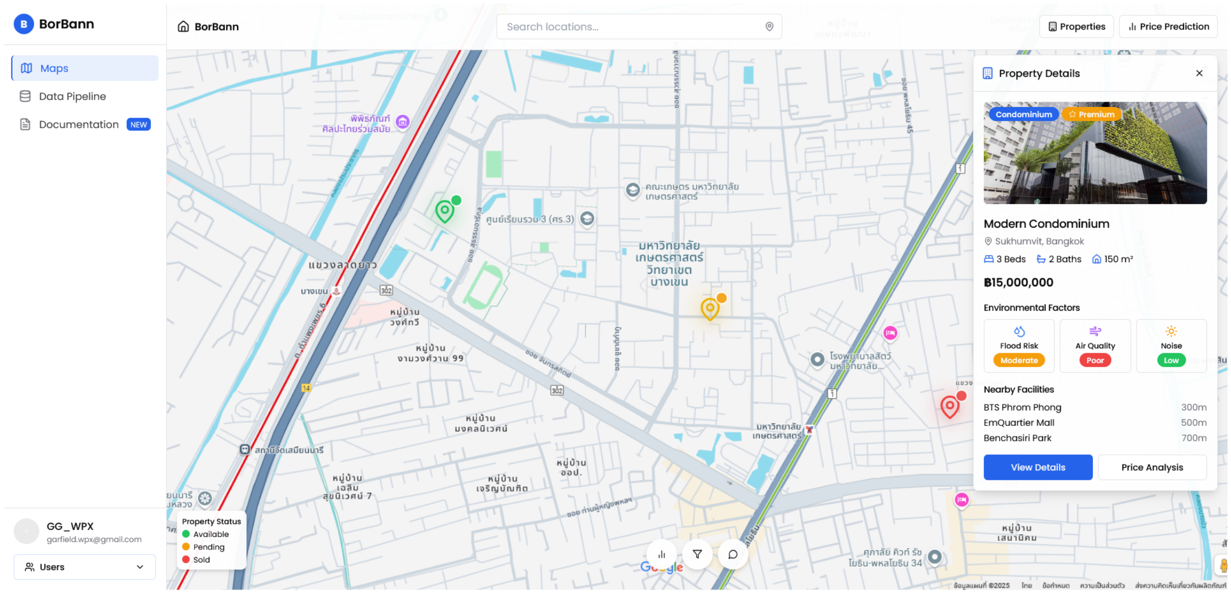


Figure 3.4: Geospatial Visualization - Main Map Interface

Figure **??** shows environmental factor visualization through heat maps and data layers. This overlay helps users analyze contextual factors like pollution levels directly on the map.

Figure **??** displays the popup that appears when users click map markers. This overlay provides immediate property information without requiring navigation away from the map.

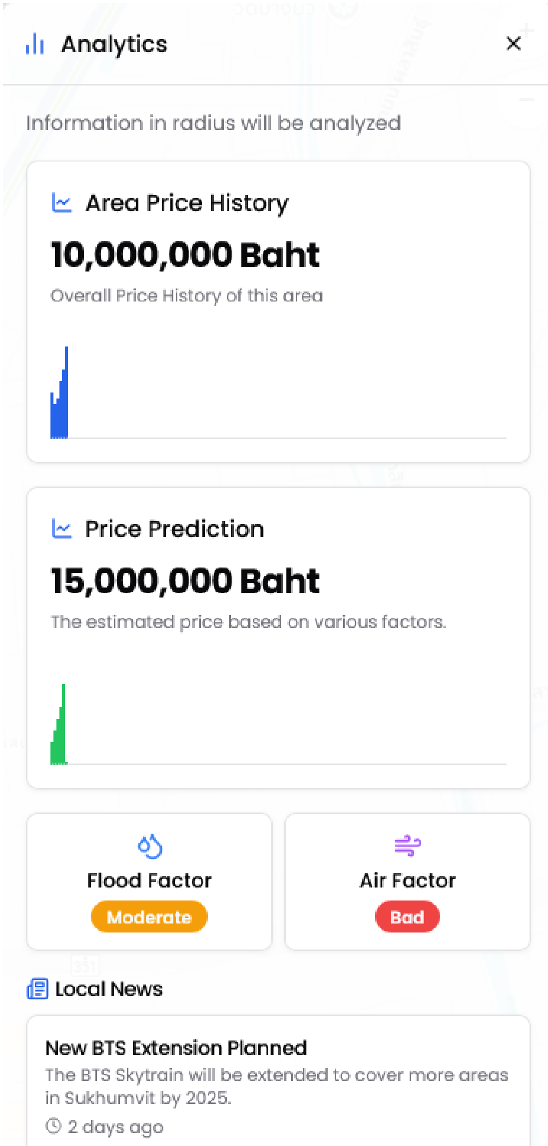


Figure 3.5: Map Analytics Overlay

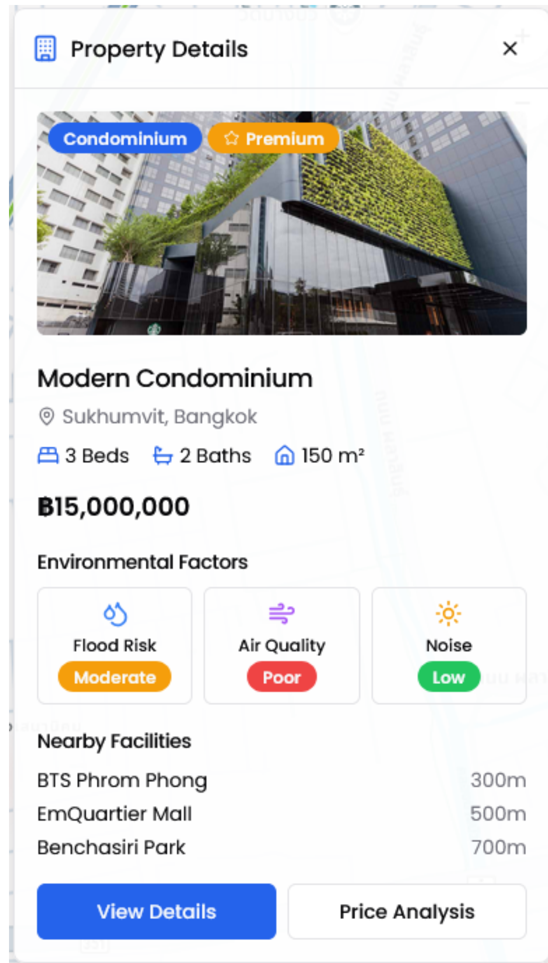


Figure 3.6: Property Detail Overlay

Figure ?? shows basic property filtering controls for price range, type and size parameters. The panel uses sliders and checkboxes for intuitive refinement of map results.

Figure ?? displays extended filtering options for specific amenities and neighborhood characteristics. These advanced parameters enable precise property matching based on detailed criteria.

Figure ?? shows the conversational assistant for property search and analysis. The chatbot handles natural language queries about properties and market conditions.

Figure ?? displays the initial analysis of property attributes in the prediction model. The interface includes interactive parameter sliders that show how changes to property characteristics affect the predicted price.

Figure ?? shows the contribution of different property features to the predicted price.

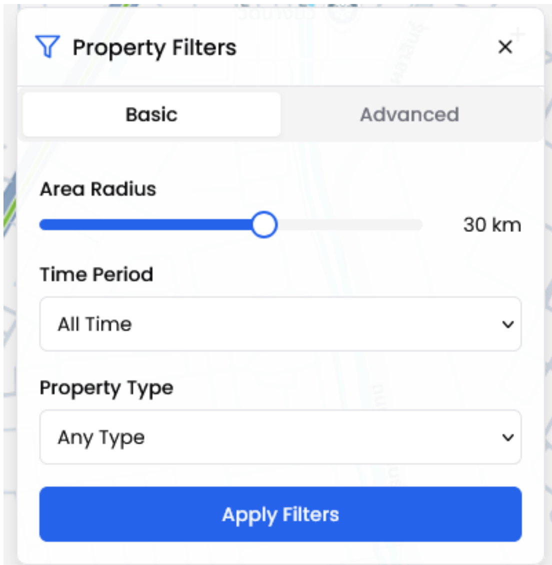


Figure 3.7: Property Filter Panel

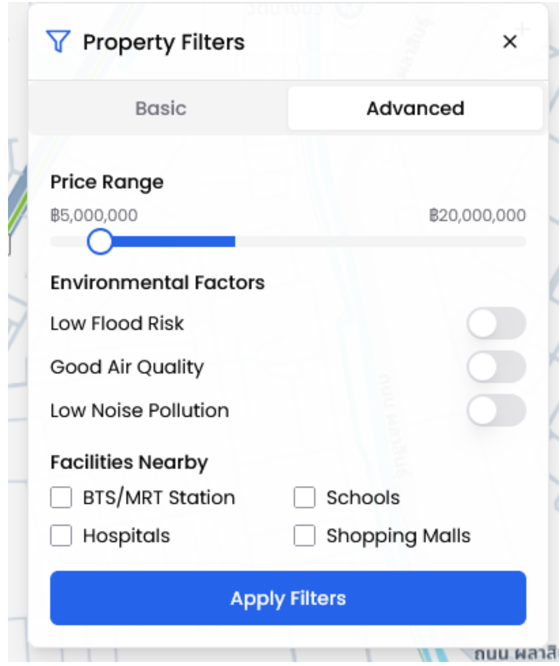


Figure 3.8: Advanced Property Filter Options

The visualization uses color-coded bars to differentiate positive and negative factors with percentage impact values.

Figure ?? presents a comparison of the subject property against similar properties in the area. The table highlights key attributes and pricing metrics with supporting analysis of market positioning.

Figure ?? analyzes environmental conditions and nearby amenities affecting property value. The interface displays flood risk, air quality, and proximity to key facilities with quantified impact on valuation.

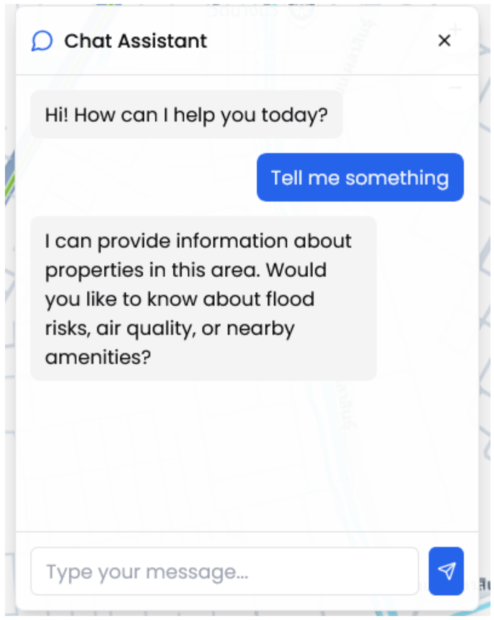


Figure 3.9: Interactive Chatbot Assistant

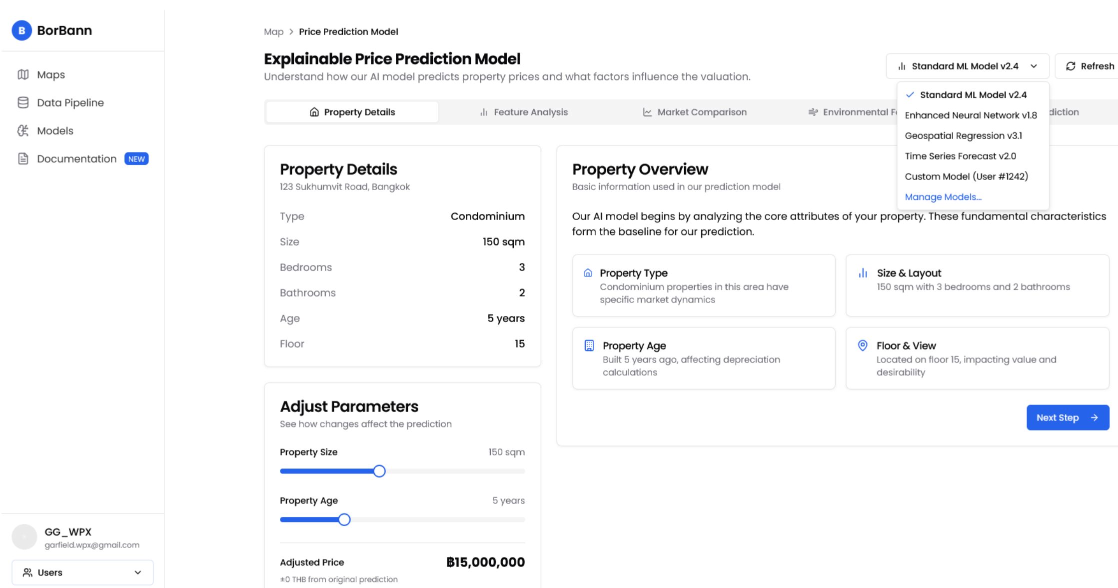


Figure 3.10: Explainable Price Prediction - Property Overview

Figure ?? shows the final price prediction with confidence metrics and range indicators. The summary section explains how the valuation synthesizes insights from all analytical components.

Figure ?? display property listings page with all listing that follow the filter tab the right.

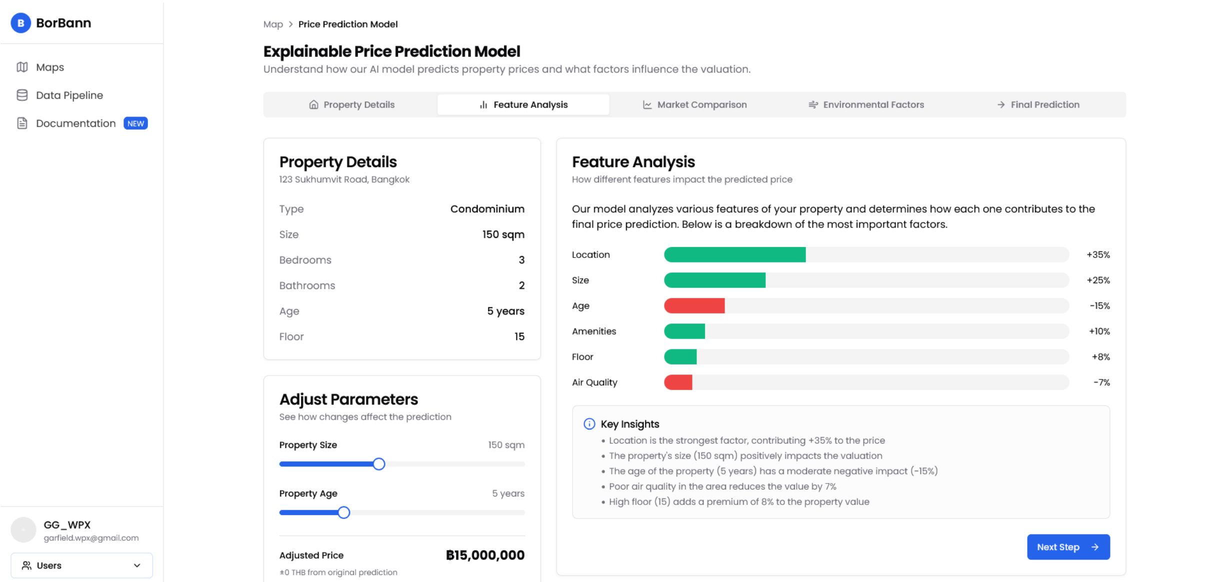


Figure 3.11: Explainable Price Prediction - Feature Importance Visualization

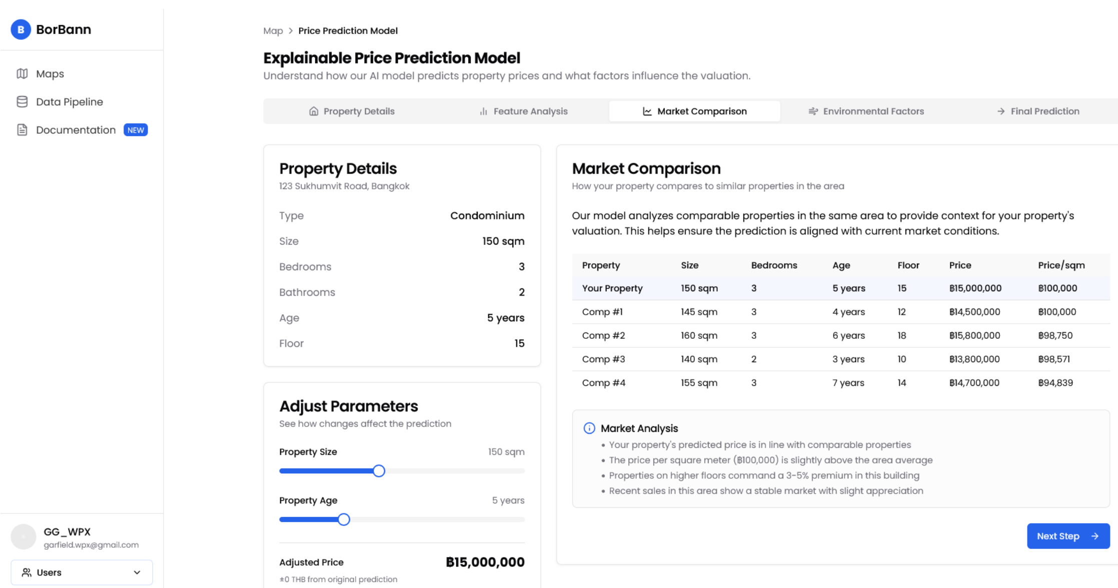


Figure 3.12: Explainable Price Prediction - Comparative Market Analysis

Figure ?? shows the analytic tab that show local context analytic of that specific listing.

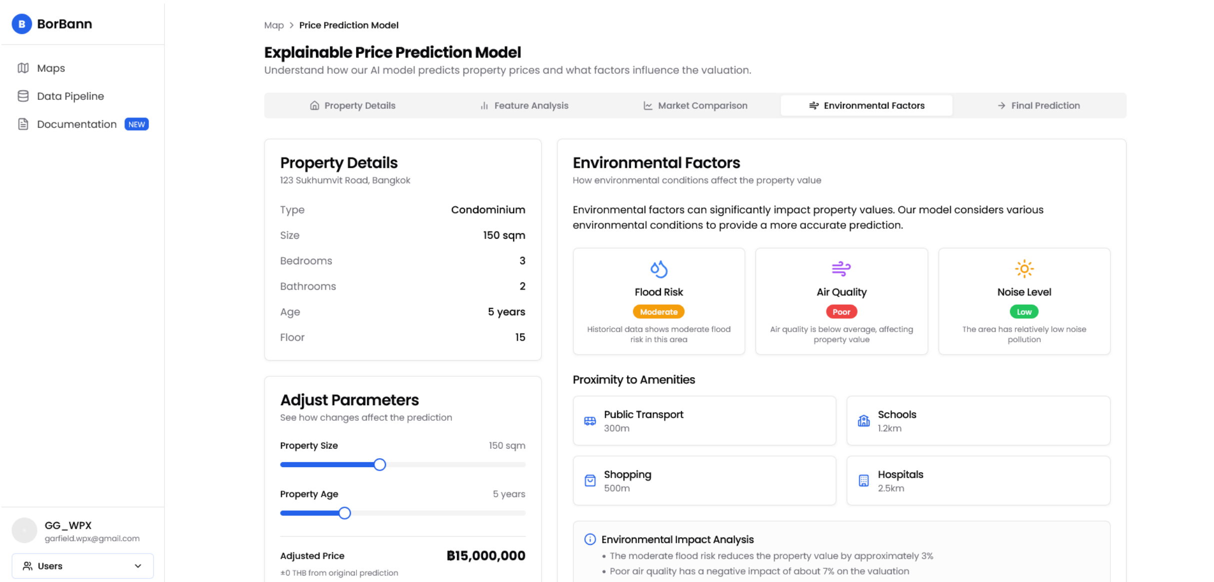


Figure 3.13: Explainable Price Prediction - Environmental Impact Analysis

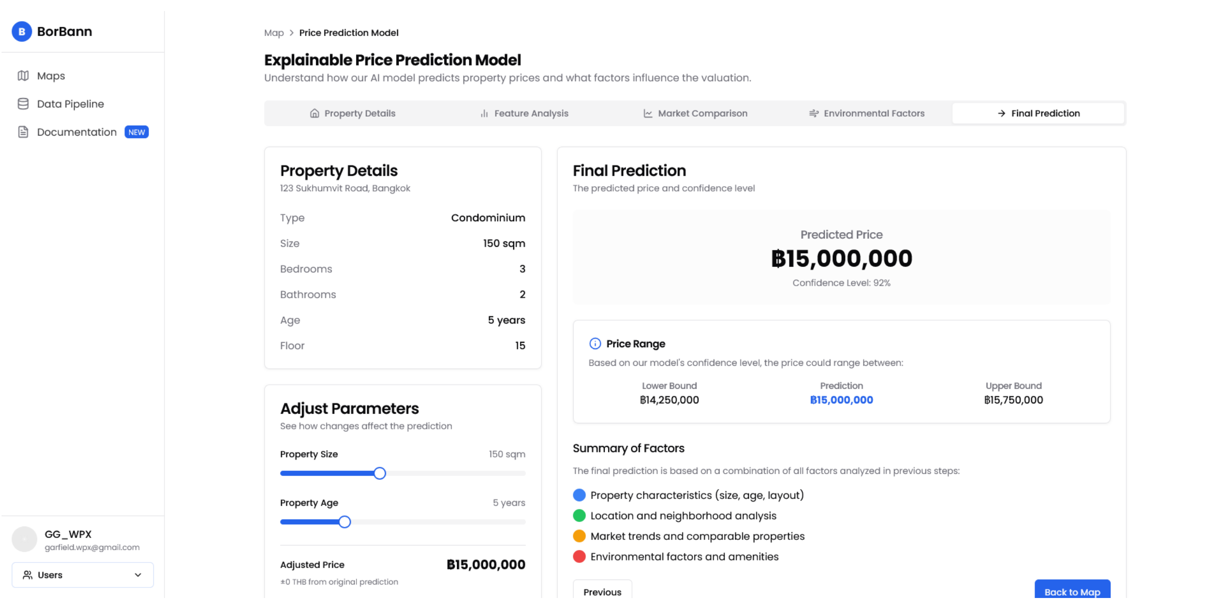


Figure 3.14: Explainable Price Prediction - Final Valuation with Confidence Range

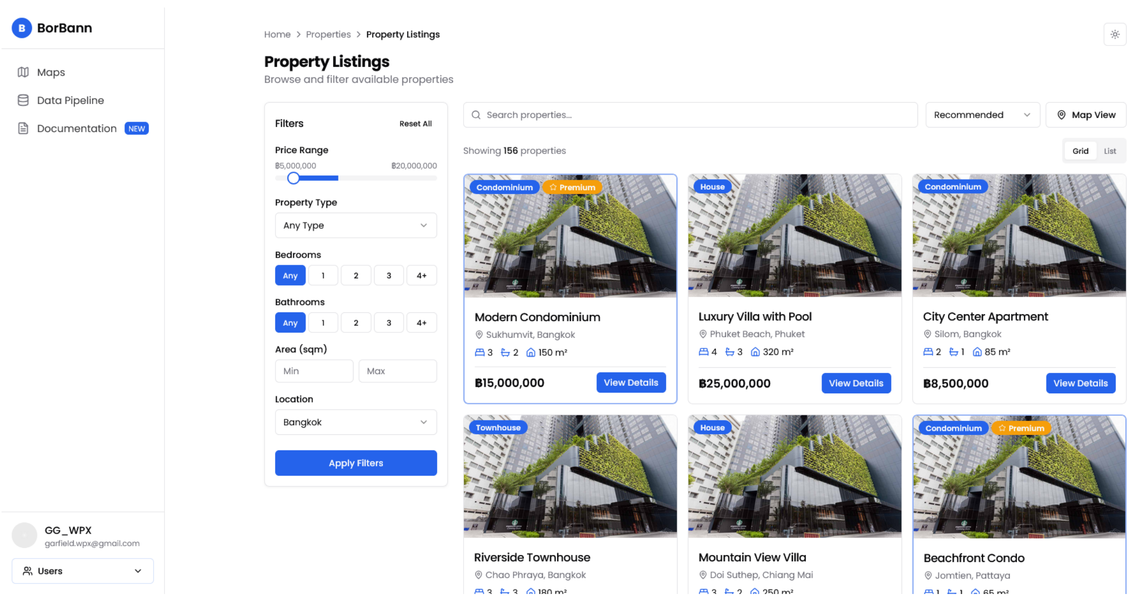


Figure 3.15: Property Listings Page

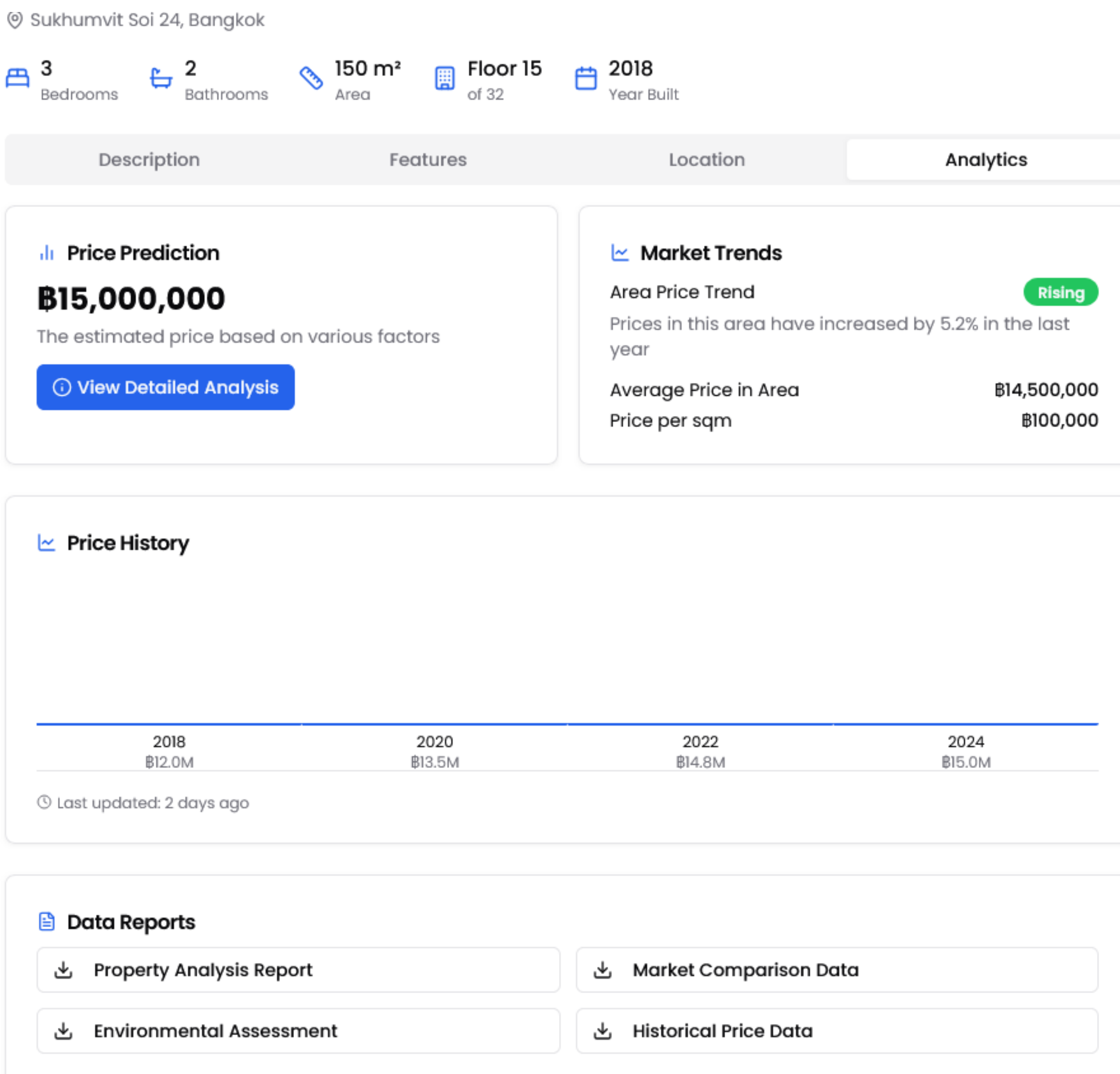


Figure 3.16: Property Listings Page - Analytic Tab

# Chapter 4

# Software Architecture Design

## 4.1 System Architecture Overview

The platform is a Retrieval-Augmented Generation (RAG) & Tool-Use Agent system specialized for real estate. It follows a "ReAct" (Reasoning + Acting) architecture where a Large Language Model (Vertex AI Gemini) orchestrates specialized geospatial tools to answer user queries.

- **Frontend:** React + TypeScript + MapLibre GL + Deck.gl (High-performance Visualization) + TanStack Query.
- **Backend:** FastAPI (Python) + LangGraph (Agent Orchestration).
- **Database:** PostgreSQL + PostGIS (Spatial Indexing) + pgvector (Semantic Search).
- **AI Model:** Google Vertex AI (LLM) + LightGBM (Production Valuation) + PyTorch Geometric (Experimental HGT).

## 4.2 Feature Implementation Details

### Borbann AI Agent (The Core)

**User Experience:** Users interact via a chat panel overlaying the map. They can attach spatial context (dropping a pin or drawing a bounding box) which is sent to the agent. The UI visualizes the agent's "thought process" by showing which tools are being called (e.g., "Searching properties...", "Calculating transit score...") before streaming the final Markdown response.

**Technical Implementation:**

- **Orchestration:** Implemented in `src/services/agent_graph.py` using LangGraph.

- **State Machine:** Uses a `prebuilt.create_react_agent` that manages the conversation history and tool execution loop.
- **Context Injection:**
  - *System Prompt:* Defines the agent's persona ("Borbann AI"), capabilities, and strict response formatting.
  - *Spatial Context:* Coordinates (lat, lon) or Bounding Boxes (bbox) from the frontend are injected into the tool arguments.
- **Tools:** The agent has access to a registry of Python functions decorated with `@tool` for automatic schema generation.

## Location Intelligence Engine

**User Experience:** Users view a "Livability Report" for any specific point, displaying scores (0-100) for Transit, Walkability, and Schools, along with Flood and Noise risk levels.

**Technical Implementation:** A weighted composite scoring system (Weights: Transit 25%, Walkability 25%, Schools 20%, Flood 15%, Noise 15%).

- **Transit Score:** Queries `transit_stops` (PostGIS) for nearest rail (¡1km), counts bus stops within 500m. Scoring uses a distance decay function.
- **Walkability Score:** Checks density of 8 amenity categories (Restaurants, Cafes, Groceries, Parks, etc.) within 800m using `view_all_pois`.
- **Schools Score:** Queries schools table within 2km, with bonuses for variety of levels.
- **Flood Risk:** Point-in-polygon check against static flood-warning datasets.
- **Caching:** Uses an in-memory dictionary cache keyed by lat/lon grid to prevent re-calculation.

## Site Analysis (Commercial Viability)

**User Experience:** Business owners select a location and a category (e.g., "Cafe"). The system outputs a "Site Score" indicating commercial potential.

**Technical Implementation:**

- **Magnets:** Counts traffic drivers (Malls, Offices, Transit) within 1km.
- **Competitors:** Counts existing businesses of the same category within 1km.
- **Heuristic Formula:** $Score = 50 + (Magnets \times 3) - (Competitors \times 2)$.
- **Traffic Potential:** Classified as High/Medium/Low based purely on Magnet count.

### Catchment Analysis (Isochrones)

**User Experience:** Displays a polygon on the map representing "How far can I walk/drive in 15 minutes?"

**Technical Implementation:**

- **Graph Engine:** Uses OSMnx to load the Bangkok road network graph.
- **Isochrone Generation:** Finds nearest graph node, traverses network based on edge weights (length/speed), and calculates Convex Hull.
- **Population Est:** User PostGIS `ST_Intersection` to overlay polygon on population grid.

### Property Search & Market Stats

**User Experience:** "Find me 3-bedroom houses in Watthana under 10MB."

**Technical Implementation:**

- **Search:** Direct translation of natural language parameters into a SQLAlchemy/PostGIS query on the `house_prices` table.
- **Market Stats:** Aggregation queries (AVG, COUNT, MIN, MAX) grouped by district.

## 4.3 Data Schema & PostGIS Usage

The system relies heavily on PostGIS for high-performance spatial queries.

| **Table** | **Geometry Type** | **Usage** |
|---|---|---|
| house_prices | POINT | Core listing data. Spatial indexing for search. |
| view_all_pois | POINT | Unified view of all amenities. |
| transit_stops | POINT | Specific transit analysis. |
| population_grid | POLYGON | 1km x 1km population density grid. |

Table 4.1: PostGIS Table Usage

# Chapter 5

# AI Component Design

This chapter describes how Artificial Intelligence (AI) components are designed and integrated into the overall system. Each section starts with a clear objective and provides guidance on analysis, design, implementation, and evaluation of AI modules.

## 5.1 Business Context and AI Integration

Figure **??** shows the overview of the system workflow. The Borbann AI Agent serves as the central orchestrator, leveraging multiple specialized engines to deliver insights. The AI integration focuses on three key components: Agentic Orchestration, Location Intelligence, and Offline Valuation.

### 1. Borbann AI Agent (ReAct)

This is the primary interface for user interaction. Using a ReAct (Reasoning + Acting) pattern, the agent interprets natural language queries ("Is this area good for a cafe?" or "Find 3-bed condos") and orchestrates the necessary tools. This allows for dynamic, context-aware responses rather than static reports.

### 2. Location Intelligence Engine

Real estate value is heavily driven by location. This engine acts as a specialized calculator, using widely recognized algorithms to score locations on:

- **Transit:** Connectivity to BTS/MRT and bus networks.
- **Walkability:** density of daily amenities (Groceries, Parks, Cafes).
- **Risks:** Flood and Noise exposure levels.

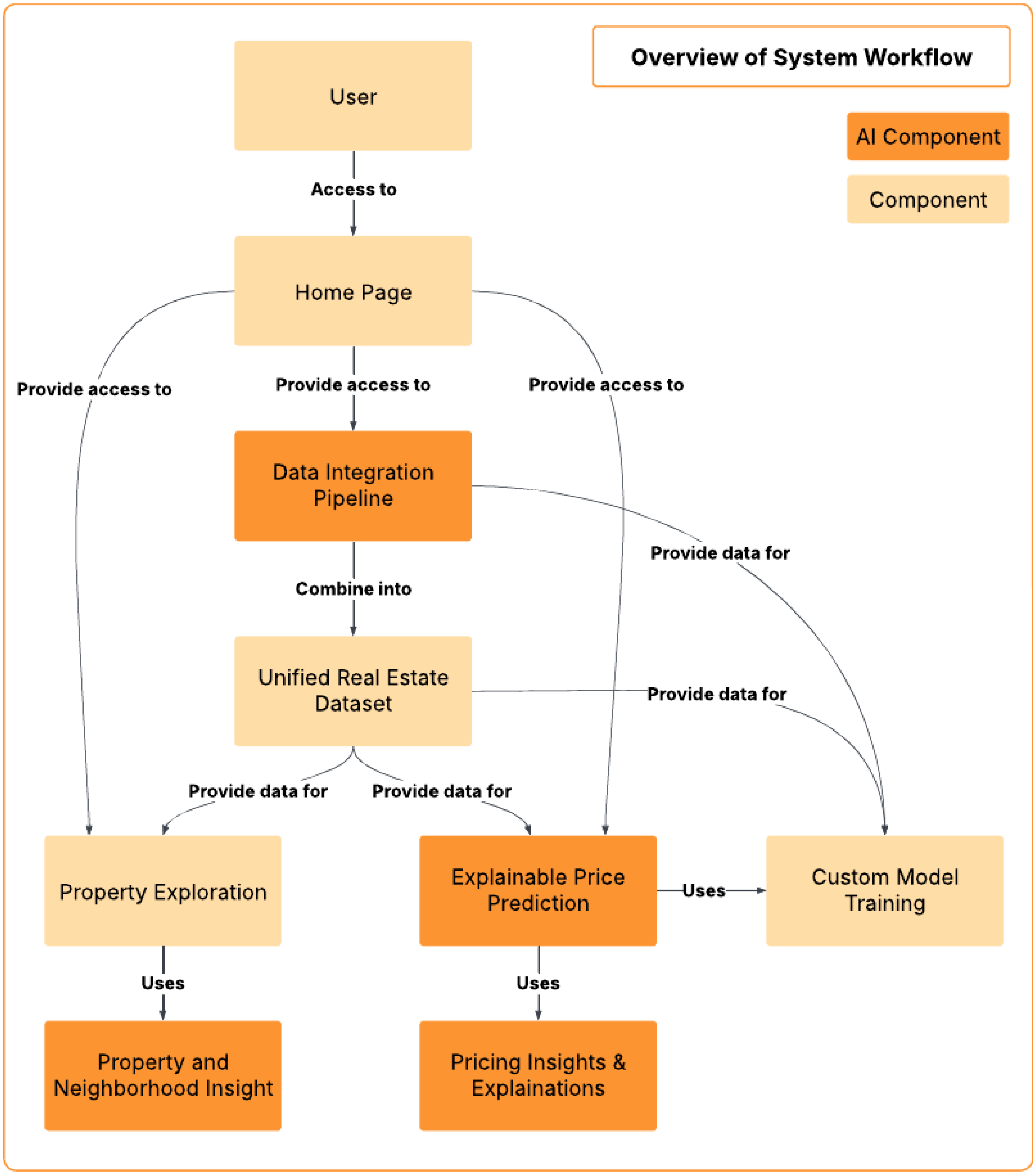


Figure 5.1: Overview system workflow (Updated)

## 3. Site Analysis & Valuation

For commercial users, the Site Analysis component evaluates business potential by analyzing "Magnets" (traffic drivers) vs. "Competitors." For residential users, the Valuation component provides price estimates using a multi-model approach. This includes a production-ready baseline model (LightGBM) enhanced with Hex2Vec embeddings for spatial context, ensuring reliable performance while advanced Graph Neural Networks (HGT) are being researched.

## 5.2 Goal Hierarchy

This section outlines the hierarchical structure of goals for our real estate valuation system.

### Organizational Goals

- To establish a trusted platform for geospatial intelligence.
- To demonstrate the viability of Agentic AI in complex decision support.

### System Goals

- To orchestrate diverse geospatial tools (PostGIS, OSMnx) via a unified Agent interface.
- To deliver sub-second responses for location scoring queries.
- To provide explainable ”Site Scores” for business decision making.

### User Goals

- To gain instant, comprehensive insights about any map location without manual research.
- To understand *why* a location is scored a certain way (e.g., ”Transit score is high because of the new MRT line”).

### AI Model Goals

- **Agent:** Correctly identify user intent and call the appropriate tool sequence 95% of the time.
- **Valuation (Hex2Vec + LightGBM):** Provide real-time price predictions with Mean Absolute Percentage Error (MAPE) below 15% in production.
- **Research (HGT):** Benchmark Heterogeneous Graph Transformer capabilities against the baseline for future deployment.

## 5.3 Task Requirements Analysis Using AI Canvas

### AI Task Requirements

#### 1. Borbann AI Agent

**Requirements (REQ):** Interpret natural language, manage conversation state, and execute tools based on spatial context (pins/bboxes). **Specifications (SPEC):** LangGraph

StateGraph with a prebuilt ReAct agent. Tools interact with PostGIS and Python service logic. **Environment (ENV):** Interactive web chat; requires handling of ambiguous user queries and potential tool execution failures.

**2. Location Intelligence Scoring**

**Requirements (REQ):** Calculate deterministic scores (0-100) for Transit, Walkability, and Schools. **Specifications (SPEC):**

- *Transit:* Distance decay function from rail stations ($< 1km$) + bus stop density.
- *Walkability:* Presence of amenity categories within 800m.
- *Flood:* Point-in-polygon lookup against flood warning zones.

**Environment (ENV):** High-performance spatial query requirement; utilizes in-memory caching for grid-based pre-calculation.

**3. Commercial Site Analysis**

**Requirements (REQ):** Evaluate the viability of a specific business type at a location. **Specifications (SPEC):** Heuristic formula: $Score = Base + (Magnets \times W_m) - (Competitors \times W_c)$. **Environment (ENV):** Dependent on up-to-date POI data in the `view_all_pois` table.

## 5.4 User Experience Design with AI

AI improves user experience by converting complex geospatial data into conversational insights and clear visualizations.

### Agent-Driven Interaction

Instead of navigating complex filter menus, users chat with "Borbann AI."

- **Scenario:** User drops a pin and asks "How is the flooding here?"
- **Response:** The agent calls the `get_flood_risk` tool and replies, "This area is in a Yellow Warning zone, historically prone to flash floods during heavy rain."

### Livability Dashboard

When selecting a location, users see a "Livability Report" card.

- **Visuals:** Radar charts showing the balance between Transit, Leisure, and Schools.
- **Interactivity:** Clicking a score (e.g., Transit) highlights the nearby stations contributing to that score on the map.

### Site Potential Analysis

For business users:

- **Input:** Select "Cafe" and click a location.
- **Output:** "Site Score: 78/100. High Potential. 3 Office Buildings nearby serves as magnets, with only 1 direct competitor within 500m."

## 5.5 Deployment Strategy

### Deployment Plan

The system is deployed as a containerized microservices architecture on Google Cloud Platform.

#### Components

- **Frontend:** Static asset serving (CDN).
- **Backend API:** FastAPI service hosting the Agent Graph and Intelligence Engine.
- **Database:** Cloud SQL for PostgreSQL (handling PostGIS and pgvector workloads).

- **AI Models:**
  - **LLM:** Vertex AI (Gemini Pro) for reasoning and tool orchestration.
  - **Production Model:** LightGBM regressor augmented with Hex2Vec spatial embeddings, served via FastAPI.
  - **Experimental Model:** Heterogeneous Graph Transformer (HGT) under active research for handling complex relationship modeling.

# Chapter 6

# Software Development

This chapter provides an overview of the software development process for the **Borbann** platform, detailing the key practices, technologies, and tools used to build and maintain the application. Effective software development is essential for creating a high-quality product that meets user needs and aligns with business goals. In this chapter, we discuss the chosen technology stack, the established coding standards, and the methodologies used throughout the development lifecycle.

## 6.1 Software Development Methodology

### Chosen Methodology

For the development of Borbann, we adopted an **Iterative and Incremental** development methodology. This approach allows the team to break down the complex requirements of a real estate AI platform into manageable components, delivering value in stages.

Key reasons for selecting this methodology include:

- **Flexibility:** Allows for adjustments based on model performance results (e.g., switching from baseline LightGBM to Graph Neural Networks).
- **Continuous Integration:** Facilitated by tools like Dagger and Docker, ensuring that code changes are regularly tested and integrated.
- **Parallel Development:** Enables the frontend and backend teams to work simultaneously on defined interfaces (APIs).

### Work Process

The work process follows a systematic approach to ensure efficiency and scalability, divided into key stages:

1. **Requirement Analysis & Feature Definition**

- Analyzing the needs for property valuation and location intelligence.
- Defining core features such as the AI Valuation Model, Chat Agent, and Interactive Map.

2. **System Design & Architecture**
   - **Database Schema:** Designing geospatial tables using PostGIS to handle complex location data effectively.
   - **API Contract:** Defining RESTful endpoints using FastAPI's Pydantic models to ensure type safety between frontend and backend.

3. **Development Phase**
   - **Backend & AI:** Developing the GIS server, training production ML models (LightGBM + Hex2Vec), and exposing them via APIs.
   - **Frontend:** Building the interactive user interface with React and integrating map visualizations using Deck.gl and MapLibre.

4. **Testing & CI/CD**
   - Automated testing using `pytest` for backend and `vitest` for frontend.
   - Continuous integration pipelines managed by Dagger to ensure build stability.

## 6.2 Technology Stack

To develop the Borbann platform, we utilized a modern, performance-oriented technology stack optimized for geospatial data and AI workloads.

### Frontend Development

- **React (v19):** A JavaScript library for building user interfaces, utilized with TypeScript for type safety.
- **Vite:** A build tool that provides a fast development environment and optimized production builds.
- **Tailwind CSS (v4):** A utility-first CSS framework for rapid and responsive UI design.
- **TanStack Query & Router:** For efficient server state management and client-side routing.
- **Deck.gl & MapLibre:** High-performance WebGL-powered libraries for rendering large-scale geospatial data and interactive maps.
- **Radix UI:** A library of unstyled, accessible components for building high-quality design systems.

## Backend Development

- **Python (3.11+):** The primary programming language, chosen for its strong support in data science and web frameworks.
- **FastAPI:** A modern, high-performance web framework for building APIs with Python, supporting asynchronous programming.
- **SQLAlchemy & GeoAlchemy2:** ORM tools for interacting with the database and handling spatial data types.
- **UV:** A fast Python package installer and resolver significantly improving dependency management speed.

## Database Management

- **PostgreSQL:** A powerful, open-source object-relational database system.
- **PostGIS:** A spatial database extender for PostgreSQL, essential for storing and querying location data (e.g., geometries, distances).

## AI & Machine Learning

- **PyTorch & PyG (Torch Geometric):** Used for developing and training Graph Neural Networks (Heterogeneous Graph Transformer).
- **LightGBM:** A gradient boosting framework used for baseline property price prediction models.
- **LangChain LangGraph:** Frameworks for building the AI Chat Agent and orchestrating LLM interactions.
- **MLflow:** Used for experiment tracking, model visioning, and managing the ML lifecycle.

## DevOps & Tools

- **Docker:** Containerization platform ensuring consistent environments across development and production.
- **Dagger:** A programmable CI/CD engine used to define and run delivery pipelines.
- **Biome:** A fast formatter and linter for the frontend codebase.
- **Make:** Automation tool used to simplify common tasks like starting the stack, running tests, and managing database migrations.

## 6.3 Coding Standards

Coding standards are enforced to ensure maintainability, readability, and consistency across the codebase.

### General Practices

- **Environment Security:** Sensitive configuration and credentials are managed via `.env` files and never committed to version control.
- **Git LFS:** Large files, particularly ML models and datasets, are managed using Git Large File Storage.

### Backend Standards (Python)

- **Type Hinting:** Comprehensive use of Python type hints to improve code clarity and enable static analysis.
- **Path Handling:** Usage of `pathlib` over `os.path` for robust, cross-platform file system operations.
- **Modular Structure:** routes, models, and business logic are separated into distinct directories (`src/routes`, `src/models`, `src/services`).
- **Package Management:** Strict usage of `uv` for adding and managing dependencies to ensure reproducible builds compared to standard pip.

### Frontend Standards (TypeScript/React)

- **Accessibility (a11y):** Strict adherence to WAI-ARIA guidelines (e.g., valid `role` attributes, meaningful `alt` text, keyboard navigability).
- **Linting & Formatting:** `Biome` is used to enforce code style and catch errors automatically.
- **Component Structure:** Functional components using Hooks, with a focus on reusability and separation of concerns.
- **Performance:** Utilization of `React.memo` and efficient state management to prevent unnecessary re-renders, especially for map components.

## 6.4 Progress Tracking

Progress throughout the development lifecycle was tracked using a combination of version control analysis and task management.

## Development Velocity

The project followed a phased implementation strategy:

1. **Phase 1: Foundation.** Setup of the PostGIS database, Dagger pipelines, and basic FastAPI structure.
2. **Phase 2: Core ML.** Data processing pipelines, training of the LightGBM baseline, and initial HGT (Heterogeneous Graph Transformer) experiments.
3. **Phase 3: Integration.** Development of the React frontend, integration of Mapbox/Deck.gl for visualization, and connection to backend APIs.
4. **Phase 4: Refinement.** UI/UX improvements, implementation of the AI Chat Agent using LangGraph, and performance optimization.

Regular commits and pull requests were used to maintain a steady stream of updates, with the 'Makefile' serving as the central hub for running development tasks and continuous integration checks.

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